

Variable separation method.

Examples

1. Using variable separation, solve Laplace's equation $\nabla^2 u = 0$ in two dimensions using plane polar coordinates,

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \phi^2} = 0,$$

and show that a solution of this equation can be constructed as

$$u(r, \phi) = C \ln r + D + \sum_{n=1}^{\infty} r^n (A_n \cos n\phi + B_n \sin n\phi) + \sum_{n=1}^{\infty} r^{-n} (\tilde{A}_n \cos n\phi + \tilde{B}_n \sin n\phi),$$

where $C, D, A_n, B_n, \tilde{A}_n$ and $\tilde{B}_n$ are arbitrary constants.

2. Consider the wave equation for the string of length l ,

(a) under gravity, $u_{tt} - c^2 u_{xx} = g$, with the ends clamped at zero: $u(0, t) = u(l, t) = 0$;

(b) without external load, $u_{tt} - c^2 u_{xx} = 0$, with a displaced end: $u(0, t) = 0, u(l, t) = h$.

Note: the equation in (a) and the boundary conditions in (b) are inhomogeneous.

In each case, seek solution in the form $u(x, t) = u_s(x) + \tilde{u}(x, t)$ and find $u_s(x)$ such that $\tilde{u}(x, t)$ satisfies the *homogeneous* wave equation with *zero* boundary conditions.

Homework problems

1. Using the form $u(x, t) = v(x)q(t)$, solve the one-dimensional wave equation,

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0,$$

for the string of length l ($0 \leq x \leq l$) with boundary conditions $u(0, t) = 0, u_x(l, t) = 0$ (i.e., fixed end at $x = 0$ and "free" end at $x = l$).

Hence, show that the string can execute harmonic vibrations described by

$$u(x, t) = A \sin \left[\pi \left(n + \frac{1}{2} \right) x / l \right] \cos(\omega_n t + \alpha),$$

with frequencies $\omega_n = \pi c(n + \frac{1}{2})/l, n = 0, 1, \dots$, and arbitrary amplitude A and phase α . [25]

2. A rope of length l and linear mass density ρ hangs freely along the x axis under gravity (acceleration g). The bottom end of the rope is at $x = 0$ and the top at $x = l$.

- (a) Using the approach used for the string, show that small displacements $u(x, t)$ of the rope satisfy the equation

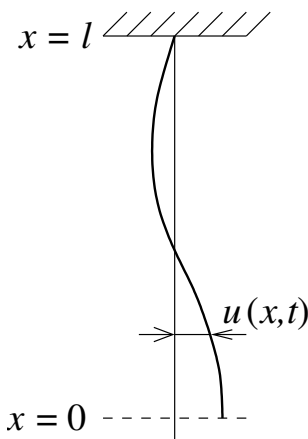
$$\frac{\partial^2 u}{\partial t^2} - g \frac{\partial}{\partial x} \left(x \frac{\partial u}{\partial x} \right) = 0. \quad (1)$$

[Hint: at point x the tension force in the rope is $T = g\rho x$.] [7]

- (b) Seeking solution of Eq. (1) in the form $u(x, t) = v(x)q(t)$, find $q(t)$ and show that $v(x)$ satisfies the equation [8]

$$x \frac{d^2 v}{dx^2} + \frac{dv}{dx} + \frac{\omega^2}{g} v = 0, \quad (2)$$

where $-\omega^2$ is the separation constant.



- (c) By introducing a new independent variable $\xi = \alpha\sqrt{x}$ (i.e., $x = \xi^2/\alpha^2$), where α is a constant, show that Eq. (2) takes the form

$$\frac{d^2v}{d\xi^2} + \frac{1}{\xi} \frac{dv}{d\xi} + v = 0, \quad (3)$$

if one chooses $\alpha = 2\omega/\sqrt{g}$.¹

[5]

- (d) Equation (3) is the Bessel equation for $m = 0$, whose regular solution is $J_0(\xi)$. Hence, show that the solutions $v(x)$ of Eq. (2) such that $v(0)$ is finite and $v(l) = 0$, are

$$v(x) = AJ_0\left(z_{0,n}\sqrt{\frac{x}{l}}\right), \quad n = 1, 2, \dots, \quad (4)$$

where A is an arbitrary constant, $z_{0,n}$ is the n th root of $J_0(z)$, and $\omega \equiv \omega_n = \frac{z_{0,n}}{2}\sqrt{\frac{g}{l}}$. [5]

- (e) Combining the results from (a)–(d), show that the rope executes harmonic vibrations with frequency ω_n , described by

$$u(x, t) = AJ_0\left(z_{0,n}\sqrt{\frac{x}{l}}\right) \cos(\omega_n t + \alpha),$$

where α is an arbitrary initial phase.

[5]

3. Consider the one-dimensional heat equation for $0 \leq x \leq l$ (rod of length l),

$$u_t - Ku_{xx} = 0. \quad (5)$$

- (a) Show that when the rod is in thermal equilibrium (i.e., the temperature does not change with time, $\partial u/\partial t = 0$), the time-independent (or *stationary*) solution of Eq. (5), $u_s(x)$, which satisfies the boundary conditions $u_s(0) = T_1$, $u_s(l) = T_2$, is

[5]

$$u_s(x) = T_1 + (T_2 - T_1)x/l. \quad (6)$$

- (b) Show that if $u_0(x, t)$ is a solution of Eq. (5) with $u_0(0, t) = u_0(l, t) = 0$, then $u(x, t) = u_0(x, t) + u_s(x)$ satisfies Eq. (5) with boundary conditions $u(0, t) = T_1$, $u(l, t) = T_2$.

[5]

- (c) Using

$$u_0(x, t) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{l} e^{-(n^2\pi^2/l^2)Kt}, \quad (7)$$

show that the solution which satisfies $u(0, t) = T_1$, $u(l, t) = T_2$ and the *initial* condition $u(x, 0) = f(x)$, is

$$u(x, t) = T_1 + (T_2 - T_1)x/l + \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{l} e^{-(n^2\pi^2/l^2)Kt}, \quad (8)$$

$$\text{where } B_n = \frac{2}{l} \int_0^l \sin \frac{n\pi x}{l} [f(x) - T_1 - (T_2 - T_1)x/l] dx. \quad [10]$$

4. Using the method of separation of variables, solve the two-dimensional wave equation in Cartesian coordinates for a rectangular membrane ($0 \leq x \leq a$, $0 \leq y \leq b$) with fixed edges, $u(0, y, t) = u(a, y, t) = u(x, 0, t) = u(x, b, t) = 0$, and show that the membrane executes harmonic motion with frequencies $\omega_{nm} = \pi c(n^2/a^2 + m^2/b^2)^{1/2}$, where $n, m = 1, 2, \dots$, and described by $u(x, y, t) = A \sin(n\pi x/a) \sin(m\pi y/b) \cos(\omega_{nm}t + \alpha)$.

[25]

¹Hint: use chain rule to transform the derivatives, $\frac{dv}{dx} = \frac{dv}{d\xi} \frac{d\xi}{dx}$, $\frac{d^2v}{dx^2} = \frac{d^2v}{d\xi^2} \left(\frac{d\xi}{dx}\right)^2 + \frac{dv}{d\xi} \frac{d^2\xi}{dx^2}$.